SLIIT
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IT1170 – Data Structures and Algorithms

Lecture 12

Heapsort

BUILD_HEAP

Input : An array A of size *n* = A.length , A.heap_size
Output : A heap of size *n*

```
BUILD_MAX_HEAP (A)

A.heap_size = A.length
for i = ⌊A.length/2⌋ downto 1
        MAX_HEAPIFY(A,i)
```

Exercise: We are given the following unordered array to build the heap.

A

1	2	3	4	5	6	7	8	9	10
4	1	3	2	16	9	10	14	8	7

Solution

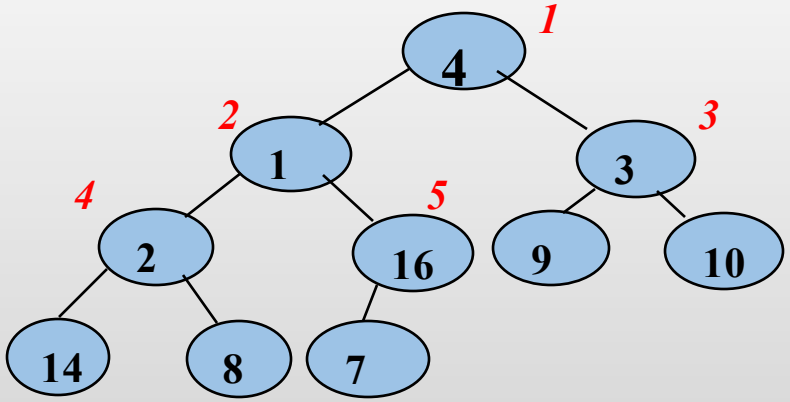
Step1

$i = \lfloor length[A]/2 \rfloor = \lfloor 10/2 \rfloor = 5$

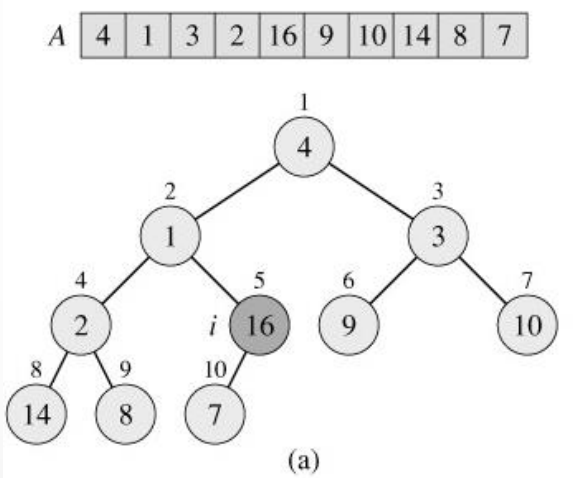
MAX_HEAPIFY(A,5)

***i* =5**

A	1	2	3	4	5	6	7	8	9	10
	4	1	3	2	**16**	9	10	14	8	7

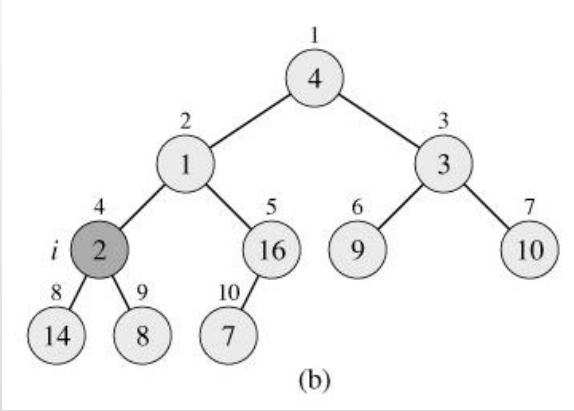


Solution (Contd.)



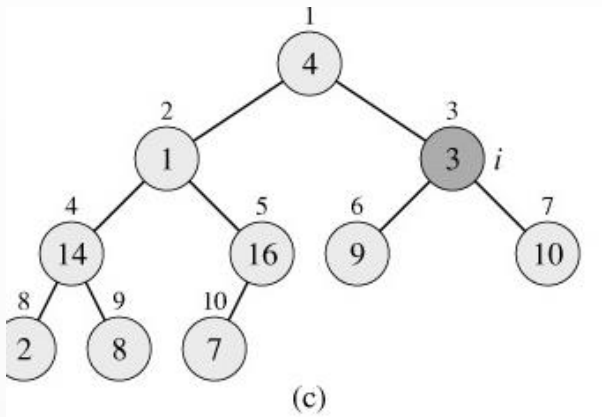
(a)

loop index *i* refers to node 5

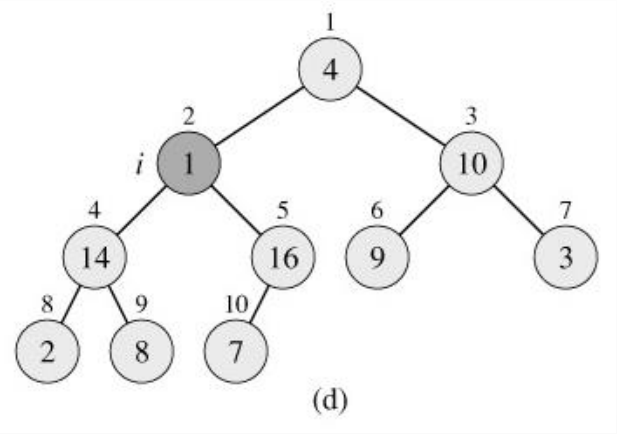


(b)

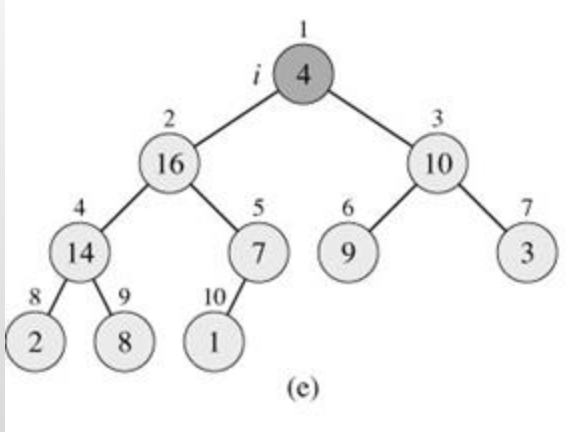
loop index *i* for the next iteration refers to node 4



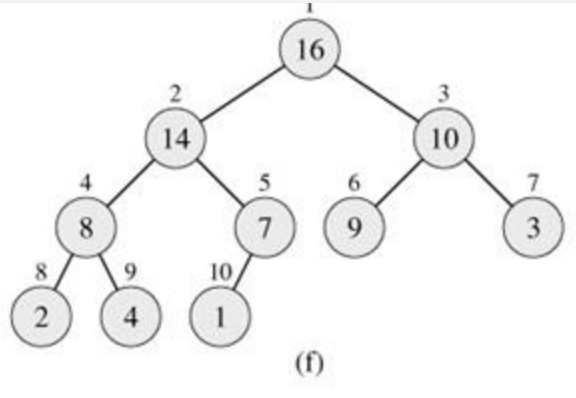
loop index *i* refers to node 3



loop index *i* refers to node 2



loop index *i* refers to node 1



max-heap after BUILD-MAX-HEAP finishes.

Analysis of Build Max Heap Algorithm.

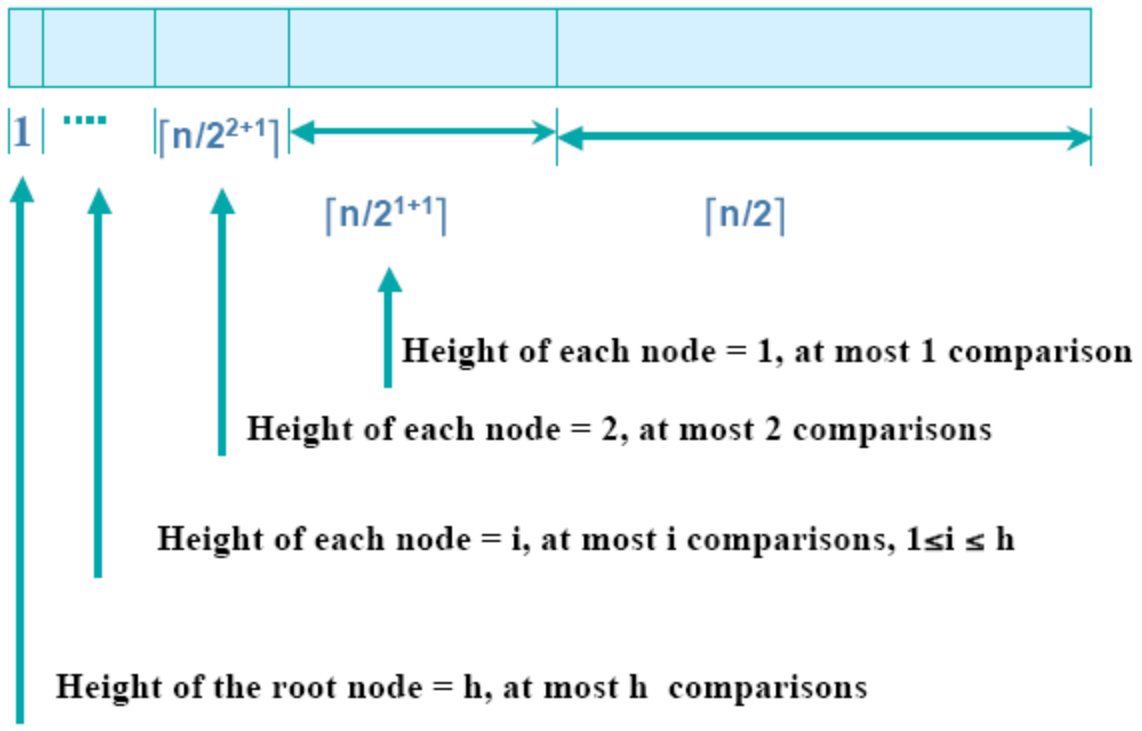
Time for MAX-HEAPIFY to run at a node varies with the height of the node in the tree, and the heights of most nodes are small.

- n-element heap has height $\lfloor \lg n \rfloor$

  and

- at most $\lceil n/2^{h+1} \rceil$ nodes of any height h.

Analysis of Build Max Heap Algorithm.



Complexity analysis of Build-Heap (1)

- For each height $0 < h \leq \lg n$, the number of nodes in the tree is at most $n/2^{h+1}$
- For each node, the amount of work is proportional to its height h, $O(h) \rightarrow n/2^{h+1} .O(h)$
- Summing over all heights, we obtain:

$$T(n) = \sum_{h=0}^{\lfloor \lg n \rfloor} \left\lceil \frac{n}{2^{h+1}} \right\rceil .O(h) = O\left(n \sum_{h=0}^{\lfloor \lg n \rfloor} \left\lceil \frac{h}{2^{h+1}} \right\rceil \right)$$

Complexity analysis of Build-Heap (2)

- We use the fact that $\sum_{k=0}^{\infty} kx^k = \frac{x}{(1-x)^2} \quad for\ |x|<1$

$$\sum_{h=0}^{\infty} \left\lceil \frac{h}{2^h} \right\rceil = \frac{1/2}{(1-1/2)^2} = 2$$

- Therefore:

$$T(n) = O\left(n \sum_{h=0}^{\lfloor \lg n \rfloor} \left\lceil \frac{h}{2^{h+1}} \right\rceil \right) = O\left(n \sum_{h=0}^{\infty} \left\lceil \frac{h}{2^h} \right\rceil \right) = O(n)$$

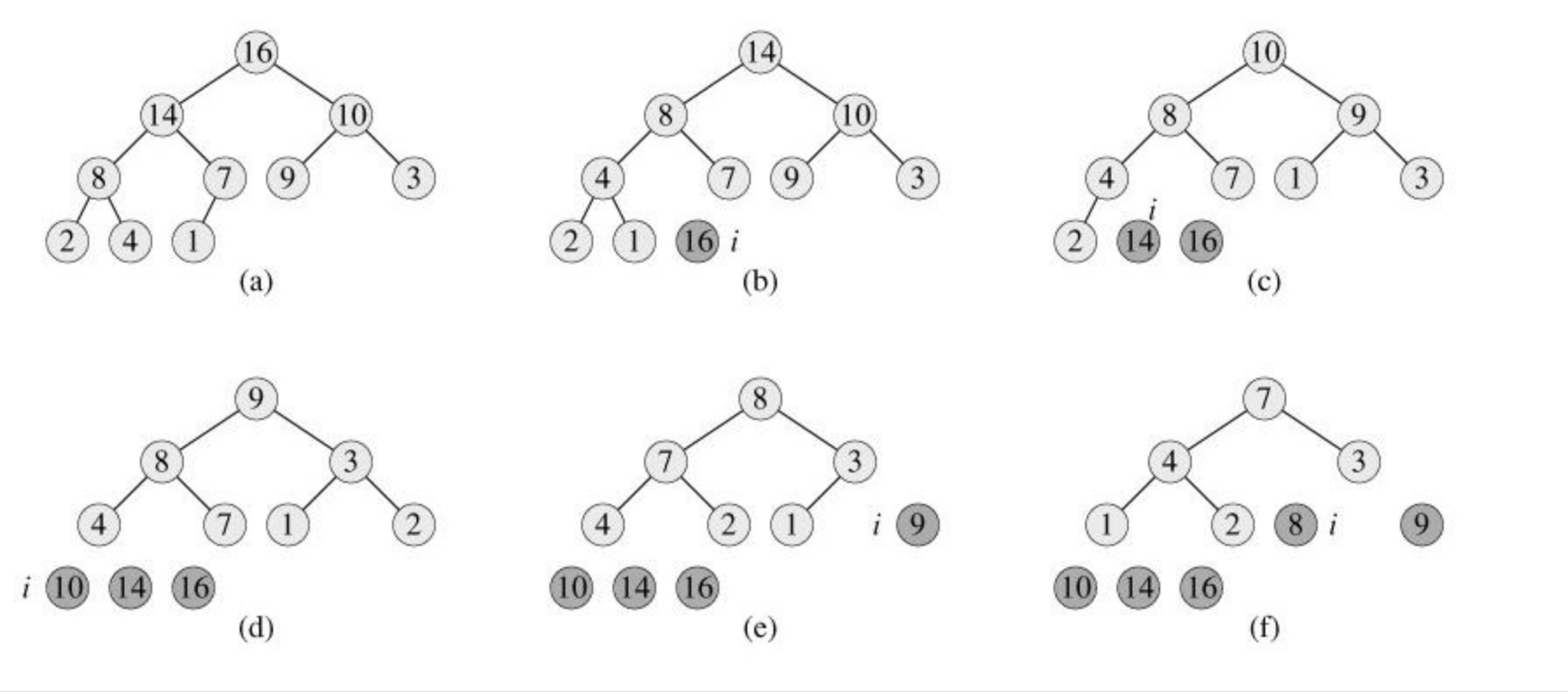
- Building a heap takes only linear time and space!

The HEAPSORT Algorithm

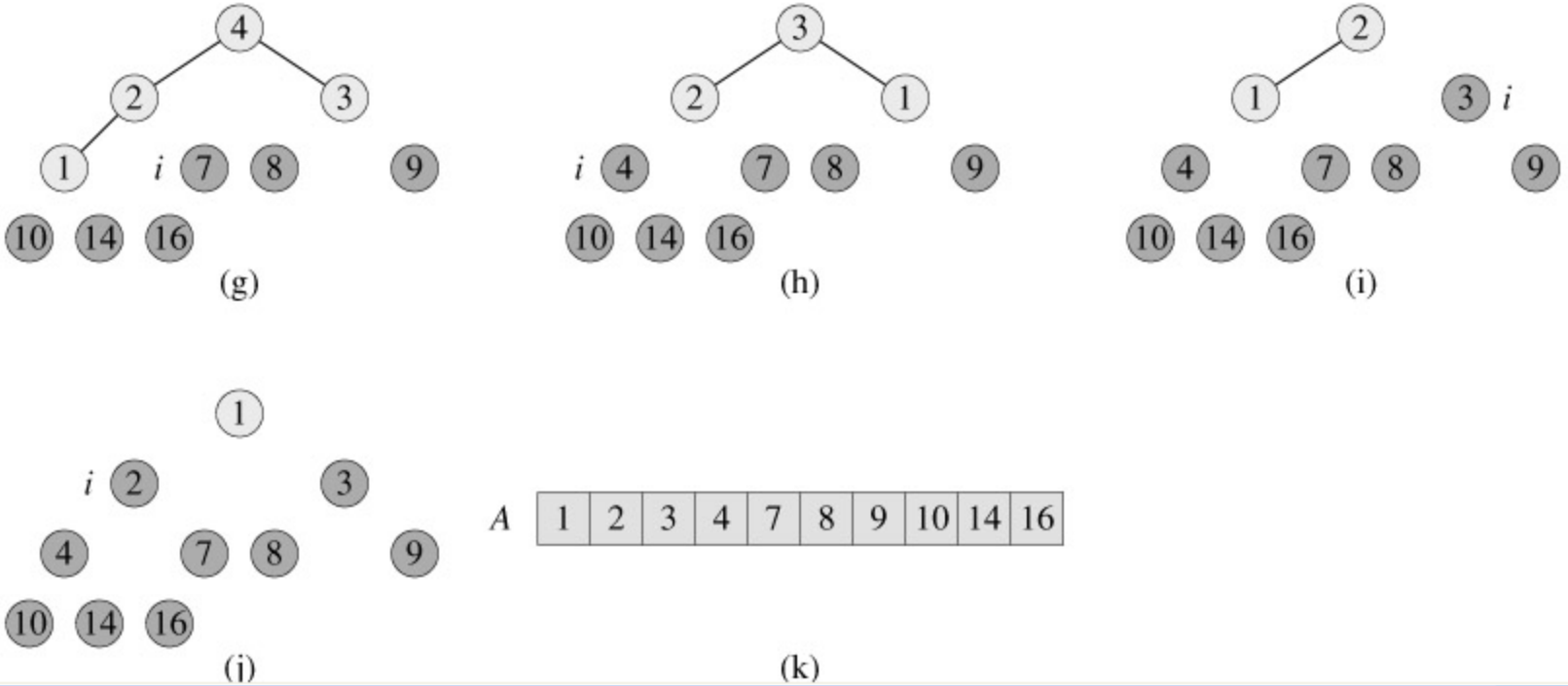
Input : Array A[1…n], n = A.length
Output : Sorted array A[1…n]

```
HEAPSORT(A)
1.      BUILD_MAX_HEAP[A]
2.      for  i = A.length down to 2
3.              exchange A[1] with A[i]
4.              A.heap_size = A.heap_size - 1;
5.              MAX_HEAPIFY(A, 1)
```

The operation of HEAPSORT.



The operation of HEAPSORT.



Heapsort Complexity

Running Time:

Step1 : BUILD_MAX_HEAP takes O(n)

Step 2 to 5 : MAX_HEAPIFY takes O(log n) and there are (n -1) calls

Running Time is O(n log n)

Priority Queues

- Heap data structure itself has many uses.
- One of the most popular applications of a heap: its use as an efficient **priority queue**.
- As with heaps, there are two kinds of priority queues:
  - max-priority queues
  - min-priority queues
- We will focus here on how to implement **max-priority queues**, which are in turn based on max-heaps

Priority queues.

- ***priority queue*** is a data structure for maintaining a set *S* of elements, each with an associated value called a ***key***. A ***max-priority queue*** supports the following operations.

- INSERT(*S*, *x*) inserts the element *x* into the set *S*. This operation could be written as $S = S \cup \{x\}$.

- EXTRACT-MAX(*S*) removes and returns the element of *S* with the largest key.

Priority queues.

- One application of max-priority queues is to schedule jobs on a shared computer.

The max-priority queue keeps track of the jobs to be performed and their relative priorities. When a job is finished or interrupted, the highest-priority job is selected from those pending using EXTRACT-MAX. A new job can be added to the queue at any time using INSERT.

HEAP_EXTRACT_MAX

HEAP_EXTRACT_MAX(A[1 .. n])

This will remove the maximum element from heap and return it

Input : heap(A)

Output : Maximum element or root, heap(A[1..n-1])

```
if A.heap_size >= 1
    max = A[1]
    A[1] = A[A.heap_size]
    A.heap_size = A.heap_size -1
    MAX_HEAPIFY(A,1)
    return max
```

Running time : O(log n)

HEAP_INSERT

HEAP_INSERT(A, key)

This will add a new element to the heap

Input : heap(A[1..n]), key - the new element

Output : heap(A[1..n+1]), with k in the heap

```
1. A.heap_size = A.heap_size + 1
2. i = A.heap_size // assume A[i] = - ∞
3. while i > 1 and A[PARENT(i)] < key
4.     A[i] = A[PARENT(i)]
5.     i = PARENT(i)
6. A[i] = key
```

Running time : O(lg n)